

Prebond Testing of Weak Defects in TSVs

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Abstract—Through-silicon vias (TSVs) are critical elements in 3-D integrated circuits susceptible to defects during fabrication and lifetime. It is desirable to detect defective TSVs in the early steps of the fabrication process to prevent stacking yield loss. Thus, the development of effective prebond testing techniques becomes of great importance. In this direction, recent research effort has been devoted to the development of two main prebond techniques: 1) prebond probing and 2) built-in self-test (BIST) techniques. The prebond probing poses economic and technological challenges, whereas current BIST proposals have disadvantages for certain solutions. Hence, there is still a need for an effective methodology in terms of fault coverage, area overhead, and test time. This paper proposes a BIST technique based on a simple unbalanced circuit comparing the behavior of two TSVs. Electrical simulation results show the viability of the proposal to detect weak defects, i.e., resistive opens and resistive bridges, adding reasonable area overhead in a short-test application time. Furthermore, an experimental design is built on a 65-nm technology, where resistive open defects are intentionally injected. Automated test equipment measurements confirm the simulation results.

Index Terms—Built-in self-test (BIST), design for testability, integrated circuit (IC) testing.

I. INTRODUCTION

3-D INTEGRATED circuits (3-D ICs) have arisen as a promising solution [1] for heterogeneous system-on-chip integration, where a stack of back-thinned 2-D ICs (tiers) is interconnected by through-silicon vias (TSVs). These TSV structures typically range between 0.4 and 200 μm for small and large sizes, respectively, with aspect ratios up to 10:1 [2]. Current fabrication techniques allow TSV densities up to 10 k/mm^2 . Therefore, these numerous structures poses a serious test challenge due to their susceptibility to defects during the fabrication, bonding to the next tier or lifetime [3]–[5].

Typical fabrication defects affecting TSVs [2]–[6] include microvoids or underfillings, which lead to weak opens. Other common defects are pinholes in the oxide, which result in undesired connections between the TSVs and the substrate, i.e., weak bridges. Ineffective removal of the seed layer may also generate shorts and/or bridges between the TSVs [5]. Misalignment during bonding may too cause defects.

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During the products service life, the differences in thermal expansion coefficients between the materials might cause thinned dies to warp, modifying the electrical characteristics of devices under physical stress [7]. Furthermore, weak opens with little or negligible impact during the normal operation may degenerate into a catastrophic defect due to electromigration [8].

Detecting defective TSVs in the early steps of the process is crucial to avoid waste of resources on completing the fabrication of circuits, which will eventually be discarded. For this reason, TSV testing is performed at different fabrication stages, divided thus into prebond and postbond testing. Whereas the postbond testing is required to ensure the quality of the final product, prebond testing, which is applied before every tier is stacked, is devoted to prevent stacking yield loss.

One of the approaches to apply a prebond test is test probing. However, TSVs are too small for test probes, and the required number of probe pads is still unaffordable. Thus, although much current research aims to overcome such disadvantages [9], [10], the prebond probing remains a challenging task. As an alternative to probe testing, built-in self-test (BIST) architectures have been developed. In the case of weak defects, the BIST circuit must consider the analog properties of the TSV. Some works have been devoted to the detection of shorts and/or bridging faults [11]–[13]. A leakage current test circuit is proposed in [11]. The test circuit measures the resistance between the TSV and the substrate, and a pass/fail signal is generated by a comparator to determine the test result. A different approach is adopted in [12] to detect the same class of faults. In this case, a resistive divider is formed, including the TSV, and the resulting TSV voltage is compared against a reference to determine the fault presence. Finally, a path-delay-based test strategy to detect short defects is proposed in [13]. Some other BIST solutions have focused the interest in detecting voids (opens) [14]–[17]. These methodologies are based on detecting a substantial modification of the parasitic capacitance associated with a faulty TSV. The works in [14]–[16] present a methodology based on a charge-sharing circuit. The TSV is initially charged and subsequently discharged. The discharge (DI) time constant of the circuit is compared against a certain predefined time thresholds to determine the presence of a fault. The advantage of [16] related to the previous works is the inclusion of a delay circuit to control the time thresholds. In [17], a capacitance bridge composed of two resistances and two capacitive elements, one of which is the TSV under test, is proposed. Finally, ring oscillators are considered in [18]–[20], where TSVs act as loads. Deviations in TSV parameters lead to variations in the propagation delay and, in turn, in the oscillation period of the ring oscillator. Such

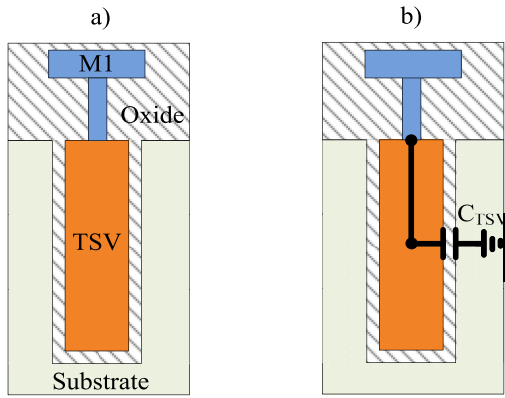


Fig. 1. TSV. (a) Structure. (b) Simplified electrical model.

techniques are applied to detect both bridging and open faults. The main improvement in [20], related to the previous work in [18], is that the BIST configuration allows simultaneous testing of multiple TSVs.

Although recent studies have improved the effectiveness of prebond testing, there is still a need for the development of methodologies with a better performance in terms of fault coverage, area overhead, and test application time. This paper proposes a BIST architecture to detect hard and weak defects by means of a simple circuit with no need to post-process test results, which adds reasonable area overhead and with short-test application time. A preliminary work on this methodology can be found in [21] and [22]. The present work adds the novelty of an enhanced BIST methodology, considering scenarios with faults affecting the TSVs pair, including a general architecture implementation and the experimental corroboration with a design built on a 65-nm technology.

The rest of this paper is organized as follows. The electrical TSV behavior during the prebond phase is briefly explained in Section II. The basis of operation of the BIST circuit is reviewed in Section III. The general BIST architecture is presented in Section IV. Section V summarizes the electrical simulation results. Section VI presents the experimental measurements on a design built on the 65-nm technology. Finally, the conclusions of this paper are discussed.

II. TSV ELECTRICAL MODELS

During the prebond phase, a TSV remains buried in silicon with only one terminal accessible through the circuitry connected to the TSV, as shown in Fig. 1(a). Although the other TSV terminal becomes exposed after wafer thinning, probing is still financially and technologically challenging.

The electrical behavior of a TSV is modeled by the RLC circuit [23]. Nevertheless, L_{TSV} becomes of concern only for high frequencies, whereas R_{TSV} is so low (in the order of milliohm) that it can be neglected in the present context. Hence, a simplified model with the predominant parasitic capacitance between the TSV and the substrate (C_{TSV}) is considered, as shown in Fig. 1(b). From here on, this is the model used to emulate the TSV electrical behavior. The typical values of C_{TSV} range between a few tens and a few hundreds of femtofarad, depending on the TSV dimensions.

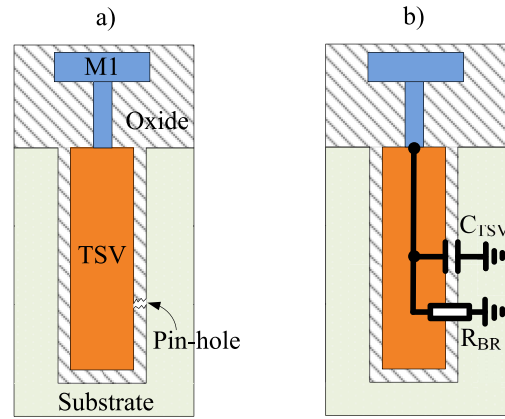


Fig. 2. Defective TSV affected by a pinhole. (a) Structure. (b) Electrical model.

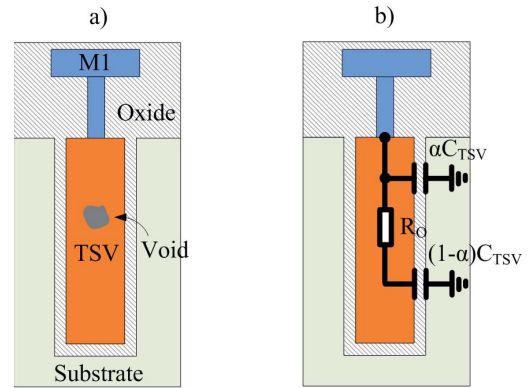


Fig. 3. Defective TSV affected by a void. (a) Structure. (b) Electrical model.

Common defects affecting TSVs lead to bridging and open faults. For example, an oxide pinhole may generate a resistive path between the TSV and the substrate, as shown in Fig. 2(a). In this case, the electrical behavior is modeled by a bridge, adding a resistance (R_{BR}) in parallel with C_{TSV} [Fig. 2(b)] [11]. In the case of incomplete metal filling, voids are generated, increasing the TSV resistance, as shown in Fig. 3(a). Therefore, the electrical behavior is modeled by adding a resistance (R_O) in series [Fig. 3(b)]. The open fault location crucially influences the behavior of the faulty TSV. This is considered in the electrical model in Fig. 3(b) [16], [20], [22]. The alpha (α) parameter accounts for the open location, ranging between 0, at the accessible terminal of the TSV, and 1, at the far end of the TSV, i.e., the floating terminal.

III. TSV BIST METHODOLOGY

The proposal to detect the TSV faults is presented here. Other considerations, such as the BIST implementation, test application time, and fault detectability, are also discussed in Sections II-B, II-C, and II-D.

A. BIST Basics

The idea consists in comparing the behavior of the TSV under test (TSV_UT) against a reference element

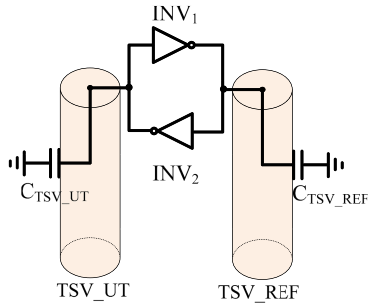


Fig. 4. Two feedback inverters connected to a TSV pair.

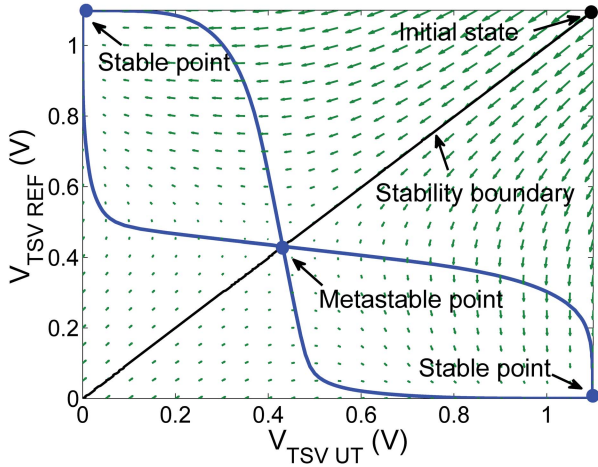


Fig. 5. Butterfly curves and phase plane [22].

with a similar electrical (capacitive) behavior, i.e., another TSV (TSV_REF). This is done by an unbalancing circuit connecting both capacitive elements [22]. This circuit is composed of two feedback inverters, as shown in Fig. 4. The behavior of the resulting system is similar to an SRAM cell in retention mode, where C_{TSV_UT} and C_{TSV_REF} represent the capacitances for TSV_UT and TSV_REF, respectively. These capacitances are expected to have similar, but not exactly the same values.

System stability is described by well-known butterfly curves, which depend on the transfer characteristic of the inverters. An example obtained from the electrical simulations from the 65-nm technology is shown in Fig. 5. The transistors have the same dimensions in both inverters. The transient behavior is determined by the associated phase plane, which depends, in turn, on the transfer characteristic of the inverters and the capacitive loads C_{TSV_UT} and C_{TSV_REF} . The system has two stable states and a metastable state. The corresponding final state is determined by the initial condition according to the phase plane and its stability boundary [24], [25]. The stability boundary has the property of not being crossed by any trajectory. It divides the phase plane into two attraction regions, one for each stable state. In this way, if the initial state set in the system is above (below) the stability boundary, the system evolves to the stable point, where V_{TSV_REF} is interpreted as a logic 1 (0) and V_{TSV_UT} as a logic 0 (1). If the initial state is situated on the stability boundary, theoretically,

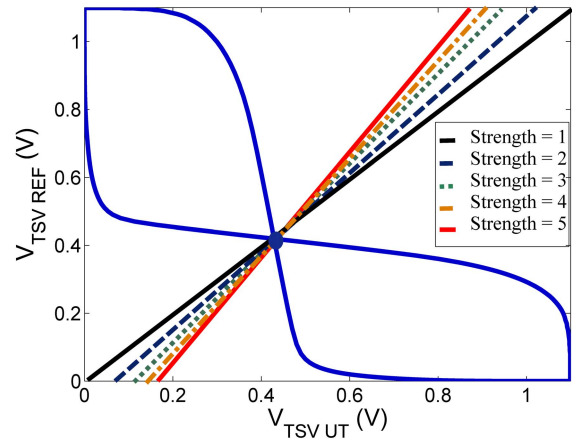
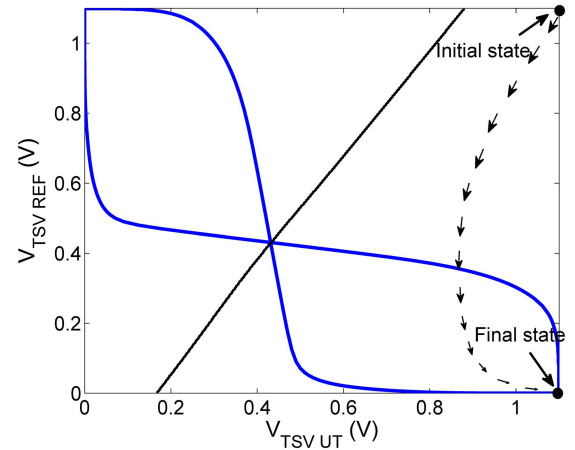
Fig. 6. Stability boundary for different relative strengths of INV₁ with respect to INV₂.

Fig. 7. Transfer characteristics and trajectory for a fault-free case [22].

the system evolves to metastability. In a real circuit, process variations and any small deviation induced by noise and/or interferences would lead the system to evolve to either of the two stable states.

Based on the circuit in Fig. 4 and given a particular fault-free design, modifying the physical parameters of the inverter transistors leads to the modification of the stability boundary. As long as the relationship between the pMOS and the nMOS transistors is kept constant, the transfer characteristics are not modified. However, different strengths between the two inverters results in a modification of the stability boundary. The simulation results in Fig. 6 show this fact, where the strength of INV₁ has been increased different times respect to INV₂. Notice how the stability boundary turns anticlockwise. With a stronger INV₁, the initial state can always evolve to the same stable state. The relative strength between the two inverters must appropriately be selected to assure always the same stable state of the fault-free circuit under process variability. An illustrative example is given in Fig. 7. The stable state represents the absence of faults, which in this particular example corresponds to V_{TSV_UT} (V_{TSV_REF}) interpreted as a logic 1 (0).

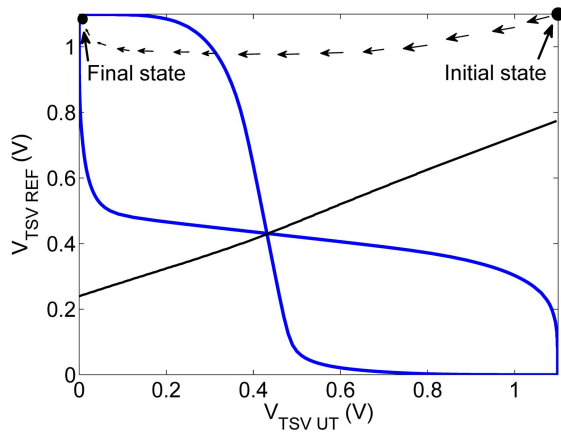


Fig. 8. Transfer characteristics and trajectory in the presence of an open fault ($R_O = 2 \text{ k}\Omega$).

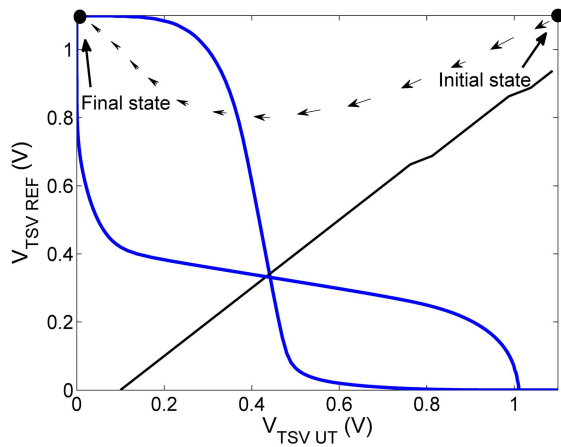


Fig. 9. Transfer characteristics and trajectory in the presence of a bridging fault ($R_{BR} = 10 \text{ k}\Omega$).

In the presence of a faulty TSV_UT, its electrical behavior is modified according to the models in Figs. 2 and 3 for the presence of a bridging and an open fault, respectively. This has, in turn, an impact on the phase plane and the stability boundary. The transfer characteristic of the inverter driving the faulty TSV is also modified by the presence of a bridging fault, but not by an open fault. However, in both scenarios, the initial state is resituated above the stability boundary, in the other attraction region, leading to a change in the final state of the system. An example for every fault type is provided in Figs. 8 and 9 in the presence of an open and a bridging fault, respectively. Notice how the transfer characteristic of the inverter driving the faulty TSV is modified by the presence of the bridge, since the pMOS transistor cannot pull up TSV_UT to V_{DD} . Anyway, in both scenarios, the faults are detected by the different stable states reached by either of the accessible terminals of the TSVs respect to the fault-free case.

B. BIST Implementation

The implementation of the proposed methodology based on the circuit in Fig. 4 is shown in Fig. 10. The stability boundary is adjusted by the different strengths of the inverters.

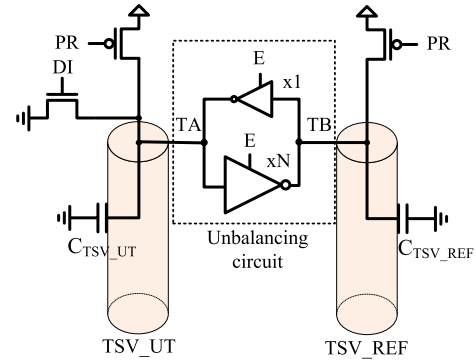


Fig. 10. Unbalancing circuit and precharge/discharge circuit.

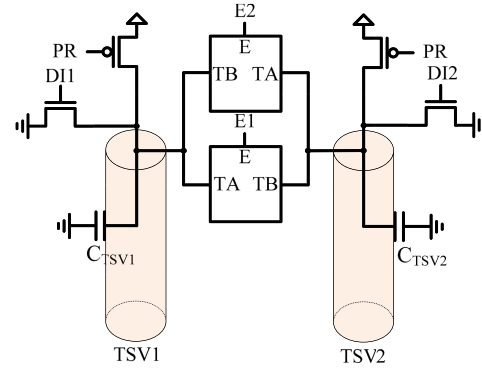


Fig. 11. BIST circuit to test a TSV pair.

The inverter driving TSV_REF is designed stronger (xN) than the one driving TSV_UT ($x1$), where $x1$ represents the reference strength. This approach leads to a stability boundary similar to that in Fig. 7, which turns anticlockwise with respect to the balanced case. The feedback inverters have been replaced by tristate inverters commanded by an enable signal (E), since during system initialization, these inverters must remain in a high-impedance state. A pull-up pMOS transistor is connected to every TSV to precharge its terminals during the initial state. These transistors are commanded by the precharge (PR) signal. A pull-down transistor is also connected to TSV_UT to generate a short discharge of the TSV to slightly modify the initial state prior to habilitation of the unbalancing circuit. The aim of this discharge is detailed in Section III-C.

The circuit in Fig. 10 can be replicated in order to detect faults also affecting TSV_REF. The general scheme is shown in Fig. 11, where the boxes represent the two unbalancing circuits. The present configuration does not specify which TSV is under test, since this depends on the control signals applied. For this reason, the TSVs have been renamed with general references as TSV1 and TSV2. The unbalancing circuit commanded by $E1$ ($E2$) and the discharge circuit, composed of a nMOS transistor and driven by $DI1$ ($DI2$), is devoted to test TSV1 (TSV2). Testing the TSV pair requires a two-step test, where one of the TSVs, i.e., TSV1, is tested on the first step, and the other one, TSV2, is tested on the second step. This fact implies activating $E1$ and $DI1$ during the first step and $E2$ and $DI2$ during the second step.

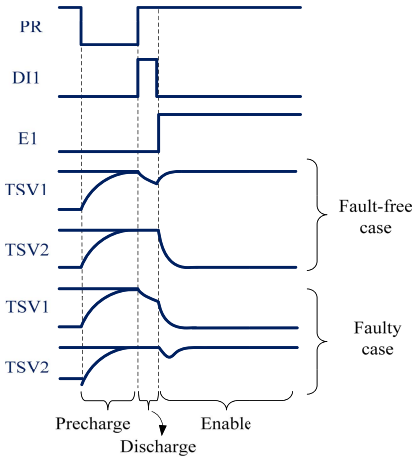


Fig. 12. Timing diagram to test TSV1.

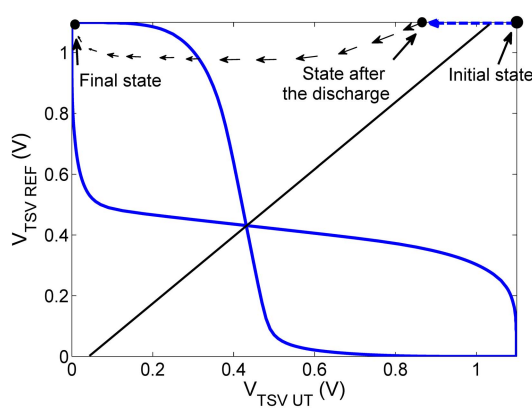
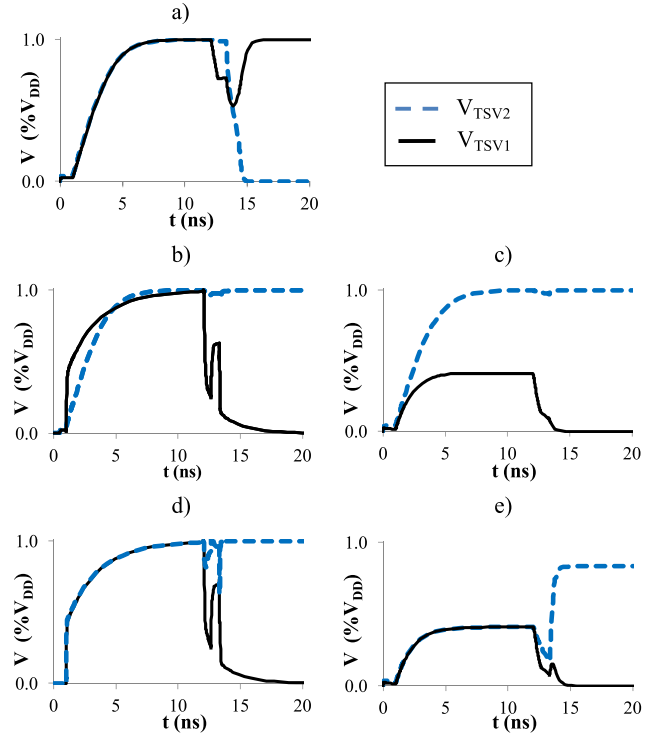


Fig. 13. Transfer characteristics and trajectory for a faulty case by applying a short discharge of TSV_UT prior to enabling the unbalancing circuit.

C. Test Procedure

Based on the BIST circuit in Fig. 11, the timing diagram to test TSV1 is described in Fig. 12. The equivalent diagram to test TSV2 can straightforwardly be derived. For the ease of understanding, E2 and DI2 are not included in this timing diagram, since they do not play any role when testing TSV1 and remain constant at logic 0. Starting from an isolated system, the first step consists in precharging both the TSVs to V_{DD} by activating the pull-up transistor. Next, a short discharge of TSV1 is performed by activating the pull-down transistor. Subsequently, the unbalancing circuit is enabled, so that the system reaches stability after a certain period of time. Then, if $TSV1 = 1$ and $TSV2 = 0$, a fault-free case is reported. Otherwise, a faulty case is found.

The short discharge of the TSV under test improves the detectability of open faults, especially when both TSVs are affected by opens with a resultant electrical behavior similar to that in the fault-free case. The short discharge is designed to have almost no influence in a fault-free TSV, since the time constant due to C_{TSV} induces a small voltage variation. However, in the presence of an open fault, the equivalent capacitance of the faulty TSV decreases, so that the discharge is faster. The initial state of the circuit is shifted to the left, being resituated in the other attraction region (Fig. 13).

Fig. 14. Transient simulation results for different scenarios. (a) Fault-free. (b) Open fault affecting TSV1 ($R_O = 10 \text{ k}\Omega$). (c) Bridging fault affecting TSV1 ($R_{BR} = 10 \text{ k}\Omega$). (d) Open fault affecting both TSVs ($R_O = 10 \text{ k}\Omega$). (e) Bridging fault affecting both TSVs ($R_{BR} = 10 \text{ k}\Omega$).

The proposed testing methodology can rapidly be applied. Assuming a typical C_{TSV} value in the order of hundreds of femtofarad and a driver with an ON resistance around kilohm, the time constant to charge/discharge a TSV remains in the order of nanoseconds. The time required to apply the whole test procedure is in the order of tens of nanoseconds at the most. For smaller TSV sizes, time constraints can be relaxed and test time shortened.

D. Fault Detectability

Different fault scenarios, including open and bridging faults, have been considered in order to demonstrate the proposal feasibility. The electrical simulation results for an illustrative example of every scenario are shown in Fig. 14, where Fig. 14(a) corresponds to the fault-free case. The precharge is applied at 1 ns and TSV1 (TSV under test) discharge at 12 ns. Subsequently, the unbalancing circuits are enabled, so that the system evolves until reaching the steady state. Notice how less than 20 ns are required for the whole test sequence. This time can be decreased for smaller TSVs, since they admit a faster precharge phase. In the fault-free scenario, the steady state, where $TSV1 = 1$ and $TSV2 = 0$, is reached. In the presence of an open fault affecting TSV1 [see Fig. 14(b)], the equivalent TSV1 capacitance decreases, so that its discharge is faster. This fact allows the system to evolve deriving the opposite steady state in comparison with the fault-free case ($TSV1 = 0$ and $TSV2 = 1$). Fig. 14(c) shows the scenario in the presence of a bridging fault between TSV1 and GND. The bridge pulls down TSV1, so that it cannot be charged,

TABLE I
TEST RESULTS FOR TSV1 UNDER DIFFERENT FAULT SCENARIOS

Fault scenario		Test result	
TSV1	TSV2	TSV1	TSV2
Fault-free	Fault-free	1	0
Open	Fault-free	0	1
Fault-free	Open	1	0
Open	Open	0	1
Bridge	Fault-free	0	1
Fault-free	Bridge	1	0
Bridge	Bridge	0	0*

evolving the system to the steady state, where $TSV1 = 0$ and $TSV2 = 1$.

As the methodology is based on the comparison between two TSVs, it may happen that, in the presence of systematic problems, both the TSVs are faulty and affected by the same fault class. In these situations, the methodology is also able to report a faulty behavior. Fig. 14(d) and (e) reports the simulation results for an open and a bridging fault affecting both the TSVs, respectively. For the double open case, TSV1 is discharged, but TSV2 remains high. Therefore, as long as the TSV1 discharge is enough to allow the subsequent evolution to the steady state, where $TSV1 = 0$ and $TSV2 = 1$, the double open scenario is detected. In the double bridge scenario, both bridges pull down the TSVs. However, as the pMOS transistor driven by TSV1 is stronger, the system evolves to the steady state, where TSV1 and TSV2 are interpreted as logic 0 and 1, respectively. Nevertheless, depending on the exact bridge resistance value, the steady-state voltages may reach degraded analog values, as observed for TSV2 in Fig. 14(e). For low resistive bridges, both voltages may be interpreted as logic 0.

The steady state reached by the system for the detected fault scenarios are summarized in Table I. It is observed how in every scenario with a fault-free TSV1, the test result is $TSV1 = 1$, whereas if TSV1 is affected by an open or bridging fault, $TSV1 = 0$. By combining the information provided by TSV2, it is possible to discriminate between a double bridge ($TSV2 = 0$) and the rest of fault scenarios ($TSV2 = 1$) when the bridge is low resistive, so that a low-voltage interpretation is also generated in TSV2. The corresponding table can easily be derived when testing TSV2.

IV. TSV BIST ARCHITECTURE

The general TSV BIST architecture presented in this section makes it possible to test multiple TSVs in parallel. The general scheme is described in Fig. 15. A BIST block, which includes the unbalancing circuits, is required for every TSV pair. This block is managed by signals TSV_S (TSV selection), PR, enable (E), and driver enable (DE). The scan register is used to scan out the results through test output (TO). The MUX connected at port D of every flip-flop (FF) selects between the functional and the test mode. Test enable (TE) control signal

is used for such purpose, so that the test mode is enabled when $TE = 1$. Although no test pattern is required, the scan register also allows the loading of a test pattern in order to test the scan register using test input (TI) or reuse it in the postbond phase. The tristate buffer driven by every FF isolates the TSVs during test. The DE signal controls this functionality, which isolates the system (logic 0) during the test, and is activated during the functional operation and/or the scan out of results.

The detail of the BIST block is described in Fig. 16(a). It comprises two unbalancing circuits, a precharge and a discharge circuit. The precharge circuit comprises two pMOS transistors, one controlled by the PR signal and the other one driven by the DE. The latter enables the precharge functionality only during the test mode and not during the normal operation. The discharge network is composed of three nMOS transistors in series. One transistor is driven by E, so that the discharge is enabled as long as $E = 0$. The corresponding TSV selection signal also commands one of the nMOS transistors to the discharge and the TSV under test. Finally, the third transistor is driven by the DI signal, which is generated internally. As the discharge time is short, the timing requirements of the input signals can be relaxed. The DI signal is generated by the pulse generator with the PR as input signal, as shown in Fig. 16(b). The delay block of the pulse generator is composed of a chain of inverters responsible for the pulsewidth, which, in turn, determines the discharge time.

Every unbalancing circuit is used to test a single TSV. The two unbalancing circuits cannot be activated simultaneously. Hence, TSV_S signal selects which TSV is under test. This functionality is controlled by means of the transmission gates (TGs) connected to enable (E) and commanded by TSV_S, as reported in Fig. 16(c). When $TSV_S = 0$, TSV1 is the TSV under test ($E1 = E$); otherwise, TSV2 is under test ($E2 = E$).

The timing diagram illustrating the test procedure with the BIST architecture is shown in Fig. 17. The TSVs are assumed to be fault-free. The test is divided into two parts. The first consists in testing every TSV1 of every pair and scanning out the results. In this way, the test mode is activated ($TE = 1$), the TSVs are set to an isolated state ($DE = 0$), and TSV1 test is selected with the TSV selection signal ($TSV_S = 0$). The testing process starts, which implies precharging the TSVs ($PR = 0$). Subsequently, a short discharge of the TSV under test (TSV1) is applied, followed by the activation of the unbalancing circuit ($E1 = 1$) to allow the system to reach the steady state. The test result is then captured by the FFs with the following active edge of the clock signal (CLK). Finally, the result is scanned out, which implies disabling $E1$ ($E1 = 0$) and enabling the tristate buffers ($DE = 1$). The second part of the test consists in testing every TSV2 of every pair. The process is similar to the above one, only that the TSV selection signal is activated ($TSV_S = 1$). Once the system reaches stability, the results are again scanned out.

In terms of area, the TGs and the pulse generator can be shared by multiple TSV pairs, reducing the area overhead. Furthermore, the BIST proposal is not expected to pose any concerns upon scaling down. In the case of smaller TSVs,

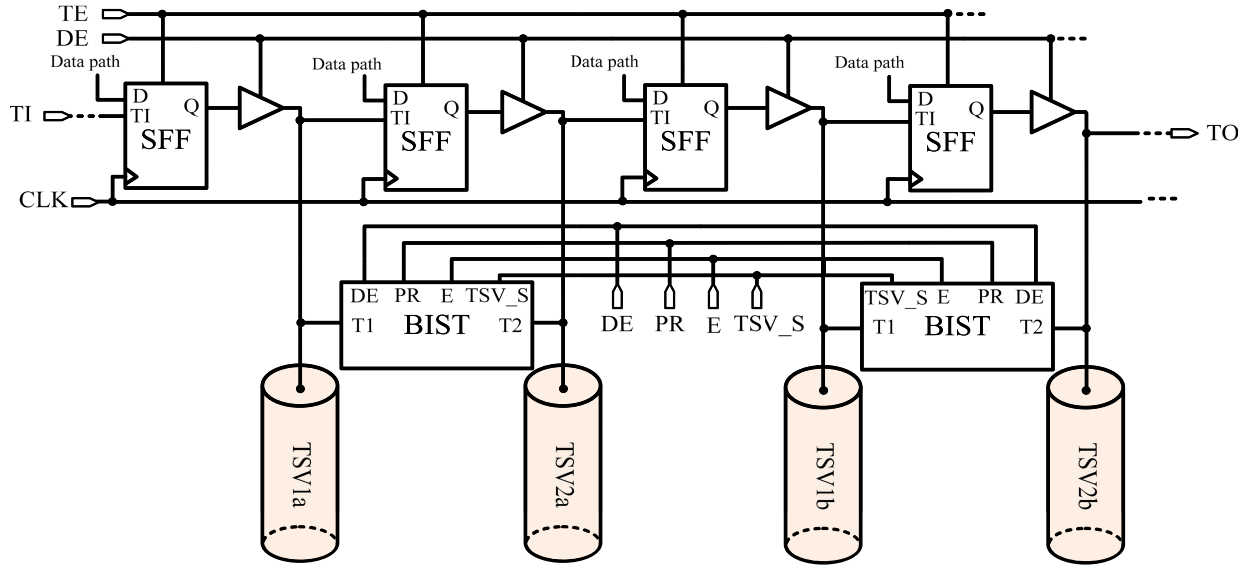


Fig. 15. BIST architecture allowing test of multiple TSVs in parallel.

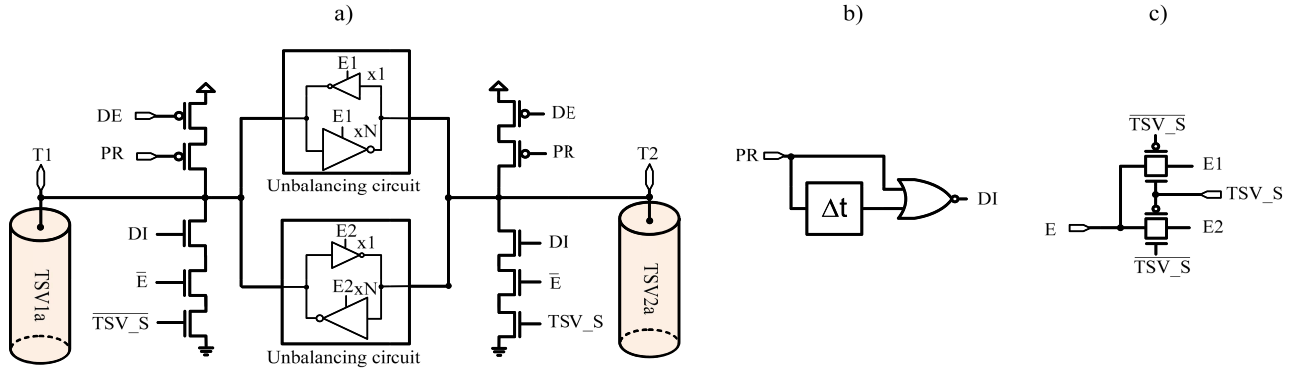


Fig. 16. Circuit detail for (a) BIST, (b) pulse generator, and (c) enable signals.

which induce lower C_{TSV} values, it allows size reduction of the unbalancing circuits as well as of the charge and discharge networks. Moreover, smaller TSVs require shorter discharge times, thus reducing the number of inverters in the delay block of the pulse generator. The BIST architecture induces simple control requirements for the inputs and output signals. Thus, it is expected that they can be accessed through an existing Joint Test Action Group (JTAG), Test Access Port (TAP) controller [26] devoted for other test purposes.

V. SIMULATION RESULTS

This section presents the SPECTRE simulation results. The circuit extracted from the BIST architecture layout was considered during the simulations based on the STMicroelectronics CMOS 65-nm technology. TSVs are emulated by their equivalent electrical models according to Figs. 1–3, depending whether they are fault-free or affected by a bridging or an open fault, respectively. A representative TSV with the following parameters is assumed: diameter = 10 μm and length = 60 μm for an equivalent C_{TSV} equal to 150 fF.

The strong inverter in the unbalancing circuit is designed four times stronger than the weak inverter. Furthermore, as the behavior of the unbalancing circuit is dominated by

the nMOS transistors, the size of the pMOS transistors is minimized in a similar way to an SRAM cell. The width of the nMOS transistor is twice that of the pMOS. The layout is designed compatible with the standard cells of the technology. The area occupied by the unbalancing, the precharge, and the discharge circuits in Fig. 16(a) is 22.8 μm^2 . Hence, 11.4 μm^2 are required for every TSV, plus the area occupied by the cells to scan out the results, the TGs and the pulse generator, which results in 20.2 μm^2 for every TSV. Therefore, the resultant area overhead is 31.6 μm^2 per TSV, which is $\sim 40\%$ of the area occupied by the assumed TSV. The possible extra area required to connect a TAP controller with the BIST architecture has not been considered. Nevertheless, this area penalty is expected to be low. The pulse generator is designed to induce a pulswidth of 0.4 ns, leading to the corresponding TSV discharge. As reported previously in Section III-B, 20 ns is sufficient to test a single TSV. Hence, the total test time is dominated by the scanout of the results, which depends, in turn, on the test clock period, which is set to 50 ns.

Fig. 18 summarizes the simulation results for open faults assuming nominal parameters. The plot shows the critical resistance (R_{O_CRIT}), i.e., the minimum open resistance detected, as a function of the open location (α) for an open

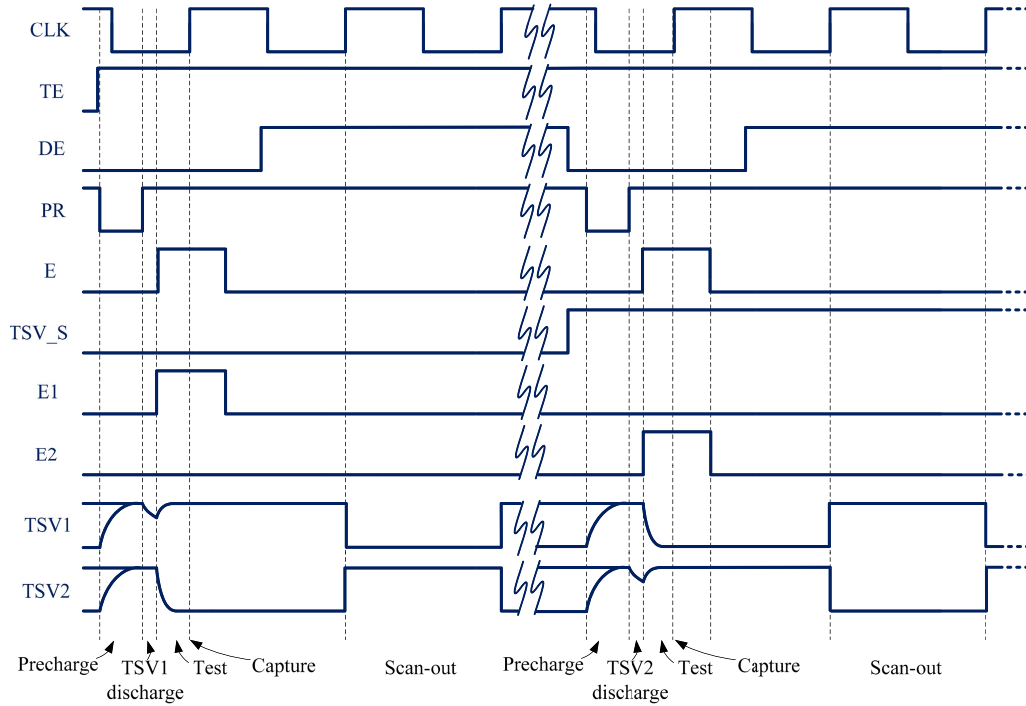


Fig. 17. Timing diagram to test TSV pairs according to the BIST architecture.

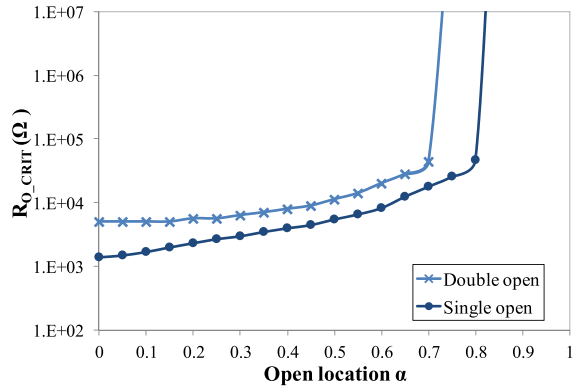


Fig. 18. Critical open resistance assuming the nominal parameters for two scenarios. Single open: open affecting the TSV under test. Double open: the same open affecting the two TSVs.

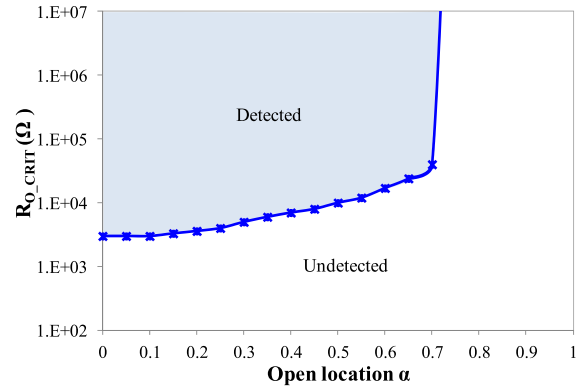


Fig. 19. Critical open resistance for an open fault affecting the TSV under test. Process variations are considered.

affecting the TSV under test (single open). A step size of 0.05 was considered for α during the simulations. At the accessible terminal of the TSV (low α values), open resistances higher than only 1.4 k Ω are detected. As the open is placed further away from the accessible terminal, the critical resistance increases approximately up to 80% of the TSV, where the fault becomes undetectable. This is expected, since the inaccessible terminal of the TSV is floating, which corresponds to the same behavior as that of a full open fault.

Monte Carlo simulations were also conducted. Variability affecting both the logic and the TSVs was considered to perform realistic simulations. In the case of C_{TSV} , variations up to 10% related to the nominal value were assumed. The results are shown in Fig. 19. The plot represents the critical resistance detected for all the runs (500 for every case).

As expected, detectability decreases with process variations so that $R_{O_CRIT} = 3$ k Ω for low α values and full opens located up to 75% of the TSV can still be detected.

In the case of systematic and persistent defects, it may happen that multiple TSVs are defective. For this reason, the simulations were performed assuming that the TSVs of the same pair were affected by an open fault. The worst scenario is found when both the TSVs are affected by an open with similar features (resistance and location) or an open with equivalent influence on the circuit behavior. The simulations were carried out assuming that an open fault with the same resistance and location was affecting the two TSVs. The results assuming the nominal parameters are shown in Fig. 18 (double open). As can be seen, detectability decreases slightly in comparison with a single open fault, with R_{O_CRIT} being ~ 5 k Ω for opens

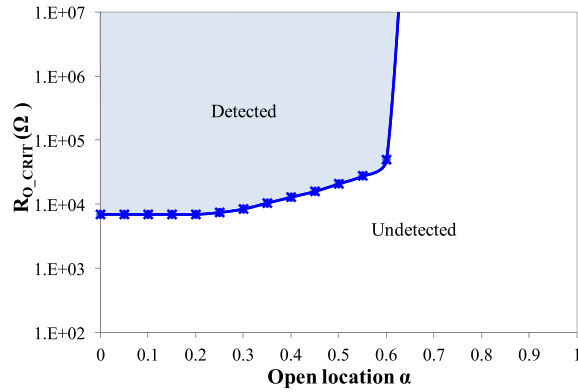


Fig. 20. Critical open resistance for an open affecting the two TSVs. Process variations are considered.

TABLE II
BRIDGING FAULT CRITICAL RESISTANCE

Fault scenario	R_{BR_CRIT} (k Ω)	
	Nominal	Monte Carlo
Single bridge	161	102
Double bridge	90	46

located at the accessible terminal of the TSV. Full opens are detected up to 75% of the TSV. The corresponding results, including process variability, are shown in Fig. 20. In this case, R_{O_CRIT} becomes 7 k Ω for low α values and full open detectability decreases to around 65% of the TSV.

The equivalent simulations were conducted for bridging faults. Fault location was neglected, because it does not have any impact on the circuit behavior. Table II summarizes the simulation results. At nominal conditions, a critical bridging resistance R_{BR_CRIT} of 161 k Ω is reported for a single bridge affecting the TSV under test, i.e., any bridging resistance lower than this value is detected by the BIST circuit. When process variability was included, R_{BR_CRIT} decreases to 102 k Ω . A double bridge affecting the TSV pair was also considered. In this case, the critical resistance is equal to 90 k Ω at the nominal conditions and decreases to 46 k Ω considering process variability.

The interconnect between the logic and the TSVs includes parasitic capacitances which may affect fault detectability. This influence can be minimized by routing the interconnect between the TSV pairs as symmetrically as possible. Nevertheless, the simulations were conducted by adding a parasitic capacitance emulating the impact of the interconnect of up to 5% of C_{TSV} . For the present technology, this scenario is equivalent to a metal line of approximately 50 μ m in length surrounded by two neighboring lines at the minimum distance allowed by the technology. This is a realistic scenario considering that the TSV diameter is 10 μ m. Even in this context, the results did not show significant differences for the configurations considered in this analysis.

We compare the present work with some of the most representative BIST methodologies found in the literature, although it must be pointed out that the conditions and technologies

assumed throughout these works are quite different. The information for this comparison is summarized in Table III. The first and second columns describe the work and the concept on which the methodologies are based, respectively. The rest of columns reports the comparison according to different criteria: 3) critical open resistance (R_{O_CRIT}); 4) range of locations for detected opens (α); 5) critical bridge resistance (R_{BR_CRIT}); 6) area overhead; 7) area overhead related to the area occupied by the considered TSV; and 8) time required to test a single TSV. Analyzing the information in Table III, the following remarks are derived.

- 1) The present work is able to detect two fault classes, unlike [11], [13], and [17].
- 2) To the best of our knowledge, the present work is the only one that considers the influence of the interconnect related with the TSVs.
- 3) Reference [17] reports better fault detectability at the expense of more area overhead. However, these results seem to be obtained neglecting C_{TSV} variability. If that is the case, including this variability will worsen the detectability of opens and bridging faults.
- 4) Reference [19] shows better open detectability but worse bridge detectability, with more area and timing penalty.
- 5) Reference [21] reports better bridge detectability at the expense of more test time. No specific information is found for the detectability of opens faults according to their location. This proposal also occupies less area, but the authors stated that they exclude some parts of the ring oscillator structure from the area computation.

In conclusion, the proposed BIST methodology reports promising results in terms of fault detectability, area overhead, and timing penalty obtained in a realistic context of simulations. Hence, it is well positioned related to the existing methodologies.

VI. EXPERIMENTAL MEASUREMENTS

A design for detectability of open defects based on the STMicroelectronics CMOS 65-nm technology was sent to fabrication to corroborate the proposed methodology. A picture of one device is shown in Fig. 21. It is a multiproject IC, where the design area and its corresponding pads occupy the bottom left area of the die, as highlighted in Fig. 21. The design is based on a similar architecture to that in Fig. 15.

Since the hit kit does not allow the use of the optional 3-D kit of the technology, the TSVs were emulated by capacitive elements. These capacitive elements are metal-insulator-metal and metal-to-metal capacitors. The capacitive behavior of a TSV during the prebond phase is expected to be similar enough to that of the emulated TSVs to derive the valuable results, which confirm the efficiency of the proposal. Moreover, the advantage of using such capacitors is the ease of injecting open defects at different locations along the capacitive element. The nominal parasitic capacitance value for these equivalent TSVs is 150 fF.

The design comprised a BIST architecture with 28 emulated TSVs pairs. Resistive open defects were intentionally injected at different locations of several emulated TSVs. Some of

TABLE III
COMPARISON OF PREBOND BIST METHODOLOGIES

BIST Proposal	Concept	R_{O_CRIT}	Locations (α) of detected opens	R_{BR_CRIT}	Area (μm^2)	Area (% TSV area)	Test time (ns)
[12]	Leakage sensor	undetected	undetected	$\sim\text{M}\Omega$	12	60	~ 50
[14]	Delay	undetected	undetected	$\sim 5\text{k}\Omega$	200	16	~ 20
[17]	Sense amplification	$> \text{k}\Omega$	[0, 0.9]	$\sim\text{M}\Omega$	200	255	$\sim$ tens
[18]	Capacitance bridge	Full open	?	undetected	48	240	~ 15
[19]	Oscillation	$> \text{k}\Omega$	[0, 0.8]	$\sim 5\text{k}\Omega$	86.4	271	~ 6000
[21]	Oscillation	$> \text{k}\Omega$	?	$\sim 3\text{k}\Omega$	10	13	~ 100
Present work	Unbalancing circuit	$> \text{k}\Omega$	[0, 0.7]	$\sim 50\text{k}\Omega$	32	40	~ 20

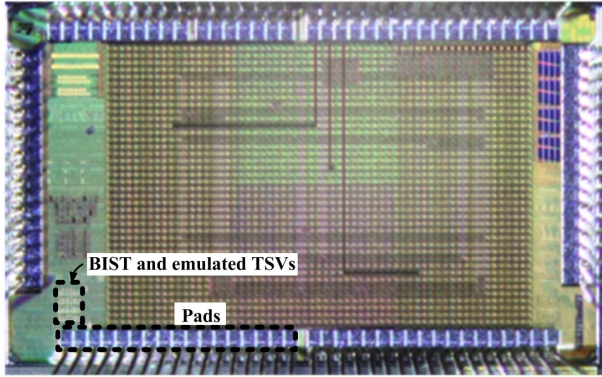


Fig. 21. Photograph of the fabricated circuit.

these opens had a fixed resistance value generated with high resistance polysilicon. Others were emulated by including a TG with full-independent controllability of the gate terminals for the nMOS and the pMOS transistors. Hence, two analog inputs were added to the design to control the gate voltage of the TGs. The interconnect between the logic and the TSVs ranged from a few tens to a few hundreds of micrometers, with intentional routing differences up to 20% of the total length. The nominal power supply voltage is 1.2 V. Among the pads pinpointed in Fig. 21, the present design uses eight inputs, one output and the power supply pads, the rest are devoted to another circuit. The inputs and the output of the present design are described as follows.

- 1) *TI (Digital Input)*: Test input used to test the functionality of the scan register.
- 2) *E (Digital Input)*: Enable of the unbalancing circuits.
- 3) *TSV_S (Digital Input)*: TSV selection signal.
- 4) *PR (Digital Input)*: TVS precharge signal.
- 5) *TE (Digital Input)*: Test enable.
- 6) *CLK (Digital Input)*: System clock.
- 7) *G_nMOS (Analog Input)*: Gate terminal for the nMOS transistor of the TGs.
- 8) *G_pMOS (Analog Input)*: Gate voltage for the pMOS transistor of the TGs.
- 9) *TO (Digital Output)*: Test output.

Measurements were performed by an automatic test equipment. The test was similar to the timing diagram in Fig. 17.

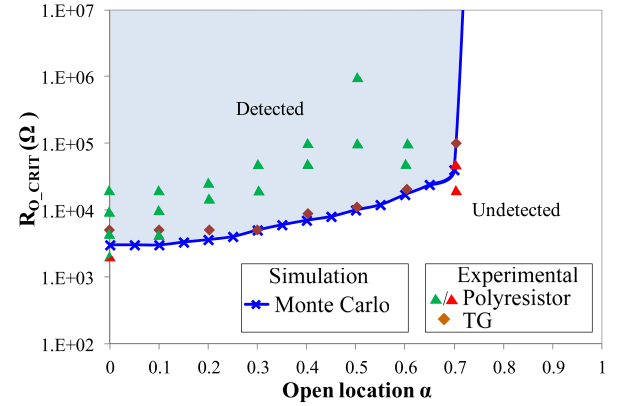


Fig. 22. Experimental measurements for open defects.

The experimental results captured by the tester are shown in Fig. 22. The simulation results are overlapped for comparison. Triangles represent open defect configurations generated with high resistance polysilicon. All the configurations are experimentally detected (plotted in green), except for three configurations, one located at $\alpha = 0$ and two at $\alpha = 0.7$. The configuration with $R_O = 2 \text{ k}\Omega$ and $\alpha = 0$ in a mixed red-green color in Fig. 22, whose detectability cannot be ensured by the simulation, is sometimes detected showing an unstable behavior. The two configurations located at $\alpha = 0.7$ (plotted in red) are not detected although the more resistive one should be detected according to the simulations results. This can be explained by two factors. The first one is that these cases are situated close to the detectability boundary or in the region, where Monte Carlo simulation results do not ensure detectability. The second is the influence of parasitic elements. In the layout design, higher routing differences than during the electrical simulations were intentionally included, which may worsen detectability for some specific configurations. Furthermore, polysilicon resistances add parasitic elements not considered during the simulations, which may also influence the detectability conditions.

The results obtained with the TG configurations are represented in Fig. 22 (brown dots). The dots are the equivalent critical resistance for the TG obtained by the simulation according to the similarity between a resistive open and

the lowest resistive configuration of the TG [$V_{g(\text{nMOS})}$ and $V_{g(\text{pMOS})}$ analog values] detected experimentally. As the TG adds parasitic capacitances and does not have a linear behavior, this critical resistance must be taken as an approximation rather than an exact resistance value. It is observed how the TG results lie close to the critical resistances obtained by the Monte Carlo simulation. Only for low alpha values ($\alpha < 0.3$) is the difference between the TG results and the critical resistance noticeable. However, in these scenarios, the TG is already ON, i.e., $V_{G(\text{nMOS})} = V_{DD}$ and $V_{G(\text{pMOS})} = 0$ V, reporting the lowest resistive configuration. Hence, lower resistances could have also been detected if the topology had allowed it. Besides, the parasitic elements of the polysilicon resistances may play a similar role for the TG configurations.

In summary, the experimental results agree with the simulation ones, confirming the feasibility of the proposal.

VII. CONCLUSION

TSVs are critical elements in 3-D ICs susceptible to defects. The detection of defective TSVs in the early steps of the fabrication process is a key factor to prevent stacking yield loss. However, current prebond techniques do not meet all demands. This paper proposes a BIST architecture to improve the detection of bridging and open defects. The solution adds reasonable area overhead in a short-test application time with no need for result postprocessing. The methodology compares the behaviors of the TSV under test with a reference TSV. Simulations carried out with the 65-nm technology show the feasibility of the proposal to detect a wide range of resistive faults. It is demonstrated that resistive bridges up to tens of kilohm are detected. On the other hand, the critical resistance for open faults, which depends on the exact fault location, is of a few kilohm for opens located at the accessible terminal of the TSV and increases with distance. Faults located at the far end of the TSV cannot be detected, since the corresponding terminal is floating. An experimental design was also built on the 65-nm technology. Open defects were intentionally injected at different locations and measurements confirm the previous simulation results and, in turn, the efficiency of the methodology.

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